Deque (Double Ended Queue)

1. Introduction

- **Deque** stands for **Double Ended Queue**.
- It is a linear data structure in which insertion and deletion operations are allowed at both ends (front and rear).
- Unlike a normal queue, elements can be inserted and deleted from both ends.
- It follows neither strict FIFO nor strict LIFO behavior.
- Deque can be implemented using:
  - Circular Array
  - Doubly Linked List
- In an array-based deque, a circular array is commonly used to utilize memory efficiently.
- Both insertion and deletion at the front and rear can be performed in **O(1)** time with a suitable implementation.

2. Working of Deque

- A deque maintains two positions:
  - **Front:** Points to the first element.
  - **Rear:** Points to the last element.
- Insertion can be performed at either front or rear.
- Deletion can be performed at either front or rear.
- In a circular array implementation, indices are updated using modulo arithmetic to wrap around the array.

Important Conditions (Circular Array):

- Next rear position: `(rear + 1) % capacity`
- Previous front position: `(front - 1 + capacity) % capacity`
- Deque is empty when `size == 0`.
- Deque is full when `size == capacity` (in a size-based implementation).

3. Basic Operations

SUNBEAM

Operation	Description	Time Complexity
`insertFront()`	Insert element at front	O(1)
`insertRear()`	Insert element at rear	O(1)
`deleteFront()`	Delete element from front	O(1)
`deleteRear()`	Delete element from rear	O(1)
`getFront()`	Return front element	O(1)
`getRear()`	Return rear element	O(1)
`isEmpty()`	Check whether deque is empty	O(1)
`isFull()`	Check whether deque is full (fixed array)	O(1)

- **Overflow:** Occurs when insertion is attempted in a full fixed-size deque.
- **Underflow:** Occurs when deletion is attempted from an empty deque.
- In Java, `ArrayDeque` provides built-in deque operations.
- Java's `ArrayDeque` does not allow `null` elements.

4. Types of Deque

1. Input Restricted Deque

- Insertion is allowed at only one end, generally the rear.
- Deletion is allowed at both front and rear.
- Restricts insertion but allows deletion from both ends.

2. Output Restricted Deque

- Insertion is allowed at both front and rear.
- Deletion is allowed at only one end, generally the front.
- Restricts deletion but allows insertion from both ends.

SUNBEAM

5. Applications of Deque

- **Palindrome Checking:** Compare characters from both ends.
- **Sliding Window Problems:** Efficiently find maximum/minimum elements in a sliding window.
- **CPU Scheduling:** Can be used in scheduling algorithms that require insertion and removal at both ends.
- **0-1 BFS:** Used to solve shortest-path problems in graphs having edge weights 0 and 1.
- **Undo/Redo Operations:** Can maintain elements or states at both ends.
- **Task Scheduling:** Supports adding and removing tasks from both ends.
- **Implementing Stack and Queue:** A deque can implement both stack and queue behavior.
- **Browser History:** Can support navigation operations in both directions.

SUNBEAM